

Supplementary Material S1

Complete Prequential Performance Across All 20 Data Streams

Dynamic Cost-Sensitive Adaptive Random Forest for Real-Time Fraud Detection in Imbalanced Data Streams

This document accompanies the main manuscript and reports the complete prequential benchmarking results (G-Mean, Precision, Recall, F_2 -Score, and total execution time) across all 20 evaluated data streams and all seven compared algorithms (Dynamic CS-ARF, ARF, HAT, OOB, UOB, CSARF-MCC, and SMOTE-Window). The main manuscript’s Table 4 presents only six primary representative streams due to space constraints; the full corpus, including all 16 synthetic drift configurations, is reported here in full. Bold values indicate the best-performing algorithm for that metric within each dataset. These are the same results used to compute the Friedman/Nemenyi statistical tests reported in the main manuscript (Section 4.3.2), and were generated with the deterministic seed (`random_seed = 42`) documented in the Reproducibility Statement. The raw, machine-readable CSV underlying this table is provided alongside this document as `Supplementary_Table_S1_full_results.csv`.

Table 1: Supplementary Table S1: Complete prequential performance across all 20 evaluated data streams and seven algorithms (G-Mean, Precision, Recall, F_2 -Score, Time in seconds). Bold indicates the best value per metric within each dataset.

Dataset	Algorithm	G-Mean	Precision	Recall	F_2 -Score	Time (s)
High Real-World Imbalance						
ULB	Dynamic CS-ARF (Proposed)	0.8644	0.5688	0.7480	0.7036	809.62
	ARF (Standard)	0.8189	0.9167	0.6707	0.7088	1452.25
	HAT	0.6880	0.6401	0.4736	0.4996	805.97
	OOB	0.7510	0.3475	0.5650	0.5022	1070.99
	UOB	0.9143	0.0970	0.8476	0.3326	372.65
	CSARF-MCC (Aguiar)	0.8300	0.9113	0.6890	0.7244	1615.71
	SMOTE-Window	0.8706	0.8776	0.7581	0.7794	1478.21
PaySim	Dynamic CS-ARF (Proposed)	0.7318	0.1159	0.5400	0.3118	181.45
	ARF (Standard)	0.3873	0.7895	0.1500	0.1790	168.73
	HAT	0.2827	0.1905	0.0800	0.0905	37.19
	OOB	0.6802	0.0571	0.4700	0.1922	142.17
	UOB	0.6520	0.0160	0.4500	0.0702	60.41
	CSARF-MCC (Aguiar)	0.3873	0.6522	0.1500	0.1773	185.95
	SMOTE-Window	0.5655	0.5246	0.3200	0.3471	184.33
Moderate Real-World Imbalance						
IEEE-CIS	Dynamic CS-ARF (Proposed)	0.5886	0.1396	0.3699	0.2781	506.43
	ARF (Standard)	0.0543	0.8000	0.0029	0.0037	851.04
	HAT	0.1357	0.3165	0.0184	0.0227	254.56
	OOB	0.6538	0.0560	0.5925	0.2032	643.94
	UOB	0.7136	0.0699	0.6817	0.2479	186.60
	CSARF-MCC (Aguiar)	0.1462	0.7436	0.0214	0.0265	942.20
	SMOTE-Window	0.3776	0.5879	0.1430	0.1685	576.95
BankSim	Dynamic CS-ARF (Proposed)	0.8710	0.4249	0.7713	0.6632	291.75
	ARF (Standard)	0.7359	0.8077	0.5426	0.5808	183.73
	HAT	0.7313	0.7743	0.5362	0.5713	46.78
	OOB	0.8467	0.1974	0.7532	0.4819	186.80
	UOB	0.8979	0.1937	0.8540	0.5078	70.92
	CSARF-MCC (Aguiar)	0.7463	0.8000	0.5581	0.5941	209.45

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Dataset	Algorithm	G-Mean	Precision	Recall	F ₂ -Score	Time (s)
	SMOTE-Window	0.8193	0.6160	0.6757	0.6629	249.41
Synthetic Drift Benchmarks						
Agrawal	Dynamic CS-ARF (Proposed)	0.8949	0.4718	0.8306	0.7210	108.49
	ARF (Standard)	0.8660	1.0000	0.7500	0.7895	42.80
	HAT	0.7366	0.9619	0.5430	0.5948	10.24
	OOB	0.9345	0.7822	0.8822	0.8602	70.34
	UOB	0.9519	0.6788	0.9231	0.8611	18.61
	CSARF-MCC (Aguiar)	0.9167	0.9940	0.8405	0.8673	47.59
	SMOTE-Window	0.9704	0.9920	0.9421	0.9517	64.70
SEA	Dynamic CS-ARF (Proposed)	0.9443	0.6823	0.8973	0.8441	62.36
	ARF (Standard)	0.8873	0.9583	0.7877	0.8168	34.37
	HAT	0.5231	0.7692	0.2740	0.3145	4.50
	OOB	0.8253	0.4477	0.6903	0.6228	53.11
	UOB	0.8782	0.1412	0.8387	0.4218	11.74
	CSARF-MCC (Aguiar)	0.8706	0.9748	0.7582	0.7934	33.61
	SMOTE-Window	0.9116	0.9021	0.8323	0.8453	33.73
Synth _{Agrawal} _{0p005_var2}	Dynamic CS-ARF (Proposed)	0.8183	0.1995	0.6923	0.4634	95.55
	ARF (Standard)	0.6666	0.9811	0.4444	0.4990	43.38
	HAT	0.5928	0.8649	0.3516	0.3990	9.95
	OOB	0.7087	0.5248	0.5048	0.5086	50.06
	UOB	0.8295	0.4519	0.6932	0.6263	17.11
	CSARF-MCC (Aguiar)	0.8584	0.9882	0.7368	0.7763	46.57
	SMOTE-Window	0.7816	0.9483	0.6111	0.6579	58.45
Synth _{Agrawal} _{0p01_var1}	Dynamic CS-ARF (Proposed)	0.9082	0.3711	0.8510	0.6761	103.05
	ARF (Standard)	0.7709	1.0000	0.5943	0.6468	47.98
	HAT	0.8194	0.9485	0.6719	0.7135	10.20
	OOB	0.8952	0.7816	0.8050	0.8002	65.55
	UOB	0.8739	0.3137	0.7921	0.6070	16.72
	CSARF-MCC (Aguiar)	0.8975	0.9701	0.8060	0.8342	47.13
	SMOTE-Window	0.9042	0.9881	0.8177	0.8469	59.23
Synth _{Agrawal} _{0p02_var3}	Dynamic CS-ARF (Proposed)	0.9264	0.4866	0.8938	0.7657	106.39
	ARF (Standard)	0.8319	1.0000	0.6921	0.7375	42.79
	HAT	0.9054	0.9935	0.8199	0.8496	10.55
	OOB	0.8903	0.6749	0.8048	0.7750	73.29
	UOB	0.9454	0.8289	0.9010	0.8856	18.88
	CSARF-MCC (Aguiar)	0.8922	0.9969	0.7960	0.8294	55.66
	SMOTE-Window	0.9347	0.9909	0.8740	0.8951	63.64
Synth _{Hyperplane} _{0p005_var2}	Dynamic CS-ARF (Proposed)	0.4043	0.0500	0.1667	0.1136	72.66
	ARF (Standard)	0.0000	0.0000	0.0000	0.0000	39.94
	HAT	0.0000	0.0000	0.0000	0.0000	6.15
	OOB	0.1368	0.0118	0.0189	0.0168	40.51
	UOB	0.4389	0.0112	0.2093	0.0462	11.96
	CSARF-MCC (Aguiar)	0.0000	0.0000	0.0000	0.0000	44.86
	SMOTE-Window	0.0000	0.0000	0.0000	0.0000	45.04
Synth _{Hyperplane} _{0p01_var1}	Dynamic CS-ARF (Proposed)	0.5381	0.0925	0.2991	0.2067	97.34
	ARF (Standard)	0.0000	0.0000	0.0000	0.0000	44.67
	HAT	0.0000	0.0000	0.0000	0.0000	7.93
	OOB	0.2289	0.0318	0.0532	0.0469	52.15
	UOB	0.6050	0.0344	0.4141	0.1292	14.13
	CSARF-MCC (Aguiar)	0.0000	0.0000	0.0000	0.0000	47.20
	SMOTE-Window	0.0949	0.1111	0.0090	0.0110	46.24
Synth _{Hyperplane} _{0p02_var3}	Dynamic CS-ARF (Proposed)	0.6621	0.1983	0.4559	0.3619	90.19
	ARF (Standard)	0.0000	0.0000	0.0000	0.0000	45.13
	HAT	0.0741	0.1111	0.0055	0.0068	6.90
	OOB	0.4353	0.1000	0.1972	0.1651	74.52

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Dataset	Algorithm	G-Mean	Precision	Recall	F ₂ -Score	Time (s)
	UOB	0.7024	0.0560	0.6045	0.2042	15.24
	CSARF-MCC (Aguiar)	0.0000	0.0000	0.0000	0.0000	50.22
	SMOTE-Window	0.2402	0.3226	0.0578	0.0692	62.85
Synth _{LED} _{0p005_var2}	Dynamic CS-ARF (Proposed)	0.6659	0.0400	0.2500	0.1220	74.31
	ARF (Standard)	0.0000	0.0000	0.0000	0.0000	85.43
	HAT	0.0000	0.0000	0.0000	0.0000	19.13
	OOB	0.6971	0.0465	0.4000	0.1587	78.84
	UOB	0.0000	0.0000	0.0000	0.0000	27.93
	CSARF-MCC (Aguiar)	0.0000	0.0000	0.0000	0.0000	85.11
	SMOTE-Window	0.0000	0.0000	0.0000	0.0000	79.84
Synth _{LED} _{0p01_var1}	Dynamic CS-ARF (Proposed)	0.6491	0.0286	0.1667	0.0847	81.17
	ARF (Standard)	0.0000	0.0000	0.0000	0.0000	80.28
	HAT	0.6363	0.0059	0.1538	0.0256	19.69
	OOB	0.7232	0.0862	0.5556	0.2660	77.09
	UOB	0.0000	0.0000	0.0000	0.0000	19.08
	CSARF-MCC (Aguiar)	0.0000	0.0000	0.0000	0.0000	86.56
	SMOTE-Window	0.0000	0.0000	0.0000	0.0000	94.93
Synth _{RandomTree} _{0p005_var2}	Dynamic CS-ARF (Proposed)	0.2745	0.0069	0.0800	0.0257	63.40
	ARF (Standard)	0.0000	0.0000	0.0000	0.0000	37.86
	HAT	0.0000	0.0000	0.0000	0.0000	5.94
	OOB	0.1373	0.0052	0.0192	0.0125	41.48
	UOB	0.3509	0.0080	0.1429	0.0325	11.06
	CSARF-MCC (Aguiar)	0.0000	0.0000	0.0000	0.0000	48.30
	SMOTE-Window	0.0000	0.0000	0.0000	0.0000	49.15
Synth _{RandomTree} _{0p01_var1}	Dynamic CS-ARF (Proposed)	0.2409	0.0095	0.0606	0.0292	84.73
	ARF (Standard)	0.0000	0.0000	0.0000	0.0000	43.09
	HAT	0.0000	0.0000	0.0000	0.0000	5.23
	OOB	0.1147	0.0037	0.0135	0.0088	52.02
	UOB	0.4432	0.0084	0.2500	0.0370	16.47
	CSARF-MCC (Aguiar)	0.0000	0.0000	0.0000	0.0000	46.49
	SMOTE-Window	0.0000	0.0000	0.0000	0.0000	47.61
Synth _{SEA} _{0p005_var2}	Dynamic CS-ARF (Proposed)	0.9686	0.8182	0.9412	0.9137	57.87
	ARF (Standard)	0.8920	0.9750	0.7959	0.8263	29.79
	HAT	0.6719	0.8841	0.4519	0.5008	4.56
	OOB	0.8604	0.5242	0.7484	0.6895	47.50
	UOB	0.8661	0.1286	0.8049	0.3922	11.16
	CSARF-MCC (Aguiar)	0.9078	0.9730	0.8244	0.8504	32.94
	SMOTE-Window	0.9209	0.9655	0.8485	0.8696	33.51
Synth _{SEA} _{0p01_var1}	Dynamic CS-ARF (Proposed)	0.9526	0.7258	0.9122	0.8676	61.89
	ARF (Standard)	0.8615	0.9703	0.7424	0.7790	32.84
	HAT	0.6310	0.8548	0.3985	0.4461	5.09
	OOB	0.8305	0.4853	0.6972	0.6412	47.43
	UOB	0.9031	0.2338	0.8514	0.5570	11.72
	CSARF-MCC (Aguiar)	0.8799	0.9360	0.7748	0.8025	35.33
	SMOTE-Window	0.9219	0.9615	0.8503	0.8705	32.90
Synth _{SEA} _{0p02_var3}	Dynamic CS-ARF (Proposed)	0.9483	0.7719	0.9053	0.8751	59.16
	ARF (Standard)	0.8545	0.9937	0.7302	0.7711	34.34
	HAT	0.6039	0.8713	0.3651	0.4131	4.18
	OOB	0.8518	0.5076	0.7378	0.6764	57.70
	UOB	0.8885	0.2410	0.8439	0.5624	11.96
	CSARF-MCC (Aguiar)	0.8794	0.9239	0.7745	0.8004	38.07
	SMOTE-Window	0.9276	0.9635	0.8612	0.8799	39.58
Synth _{Waveform} _{0p02_var3}	Dynamic CS-ARF (Proposed)	0.6943	0.1575	0.4554	0.3305	77.15
	ARF (Standard)	0.2478	1.0000	0.0189	0.0235	64.86
	HAT	0.5954	0.0790	0.3500	0.2076	18.00

Synth_{Waveform}_{0p02_var3}

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Dataset	Algorithm	G-Mean	Precision	Recall	F ₂ -Score	Time (s)
	OOB	0.6663	0.0802	0.4597	0.2361	79.86
	UOB	0.6474	0.1429	0.3832	0.2867	20.98
	CSARF-MCC (Aguiar)	0.0000	0.0000	0.0000	0.0000	57.54
	SMOTE-Window	0.4001	0.5714	0.0792	0.0957	57.26